Beyond Dodge Rate: Testing Humanoid Dodgeball Under Camera Failures

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https://hebbian-robotics.github.io/beyond-dodge-rate/

***Abstract*—Dodge rate measures whether a robot avoids an incoming ball, but can miss later task failure. We study this evaluation gap in simulated humanoid dodgeball, where task success also requires staying upright and keeping the pelvis within 1 m of its starting position throughout a four-second trial. Controlled camera interventions test two fixed G1 policies. In a paired 64-seed experiment, restoring fresh frames at 2 s raises π_1 success from 50.0% to 95.3%, while dodge rate remains 100%. In a separate stale-to-blank control, blank input and fresh-image restoration rescue the same 31 of 64 trials for π_1 and leave π_2 success unchanged. Yet fresh frames bring both policies closer to their starting position by trial end than blank input does. These simulation findings support reporting post-encounter motion alongside contact, falls, and area compliance.**

I. Introduction

A humanoid can dodge a ball and still fail its task by falling or leaving the allowed area (Fig. 1). Interpreting dodge rate as task success can therefore hide failures after the encounter. Evaluation must cover the whole trial.

PAC-MAN evaluates contact and falls, including return to station between throws [13]. Blackout vs. Freeze shows that similar task success can conceal different physical consequences of camera faults in manipulation [12]. We examine this evaluation gap in humanoid evasion. Paired camera interventions expose two distinctions: unchanged dodge rates can accompany different area-compliance outcomes, and identical success rates can accompany different final pelvis distances. A stale-to-blank control asks whether the observed benefit requires fresh images or also follows removal of the stale frame. Shared throws and post-hoc scoring analyses test dependence on the throw distribution and area radius.

II. Methods

A. Policies, task, and camera input

We use Russi's G1 Depth Dodge code [9], based on ViBe/ORCS [6] and LeRobot G1 assets [5]. It combines frozen SONIC [7] and Theia [11] bases with depth adaptation trained by PPO [10]. π_1 is the released iteration-4000 model [1]; π_2 was trained separately with the same method, architecture, and recorded settings. Their incomplete training histories prevent attributing differences to a training factor. Evaluation fixes both policies and native control settings. All trials correct an inactive-ball collision bug by disabling collisions while the ball is parked below the floor and restoring them at launch.

The G1 must dodge a ball, stay upright, and keep its pelvis inside an allowed area of radius 1 m centred on the pelvis starting position. The ball launches at 1 s; trials last 4 s or until a fall. The simulator aims at one of six body locations, adds random position offsets and $\pm 30^\circ$ direction variation, and uses a nominal flight time of 0.55–0.62 s.

Healthy input supplies fresh depth frames. Stale input repeats the last pre-fault frame, and blank input supplies zero-valued frames. Faults replace frames before encoding. Theia encodes frames, and a perception module aggregates 16 visual tokens. The actor combines this feature with a standing reference and ten-step histories of gyroscope readings, joint-position offsets, joint velocities, previous actions, and projected gravity. It produces 29 normalized joint actions for native PD control. Histories keep updating through faults and switches.

B. Paired experiments

The input-handling experiment compares healthy, stale, and blank input to test how faults change dodge rate and area compliance. The fresh-image restoration experiment tests whether resuming fresh frames changes post-encounter motion. The restoration timeline is

$$\text{launch } 1 \text{ s} \rightarrow \text{fault } 1.16 \text{ s} \rightarrow \text{camera switch } 2 \text{ s} \rightarrow \text{end } 4 \text{ s}$$

The camera switch follows this fixed schedule rather than being triggered by throw resolution. Histories keep updating and are not reset. All stale-input throws resolve by restoration within numerical tolerance. After a separate pilot, we fixed 64 new seeds and the primary contrast: restoration versus continuous stale input for π_1. The other contrasts are secondary.

Within each policy and seed, we match prelaunch robot positions and velocities, launch time, target, ball position and velocity, and flight time across camera conditions. We also verify camera schedules. Restoration pairs match recorded physical traces before 2 s. Missing or mismatched trials halt analysis. Different prelaunch motion can change the simulator's aim, so equal seeds need not give equal throws across policies.

The exploratory shared-throw control tests whether the response differences persist when both policies face the same ball launches. It uses 64 recorded launches, taking 32 from each policy's healthy-input trials without selecting for outcome. Both policies receive each initial ball position and velocity under all five camera conditions. Robot states can still differ, and subsequent ball trajectories are simulated rather

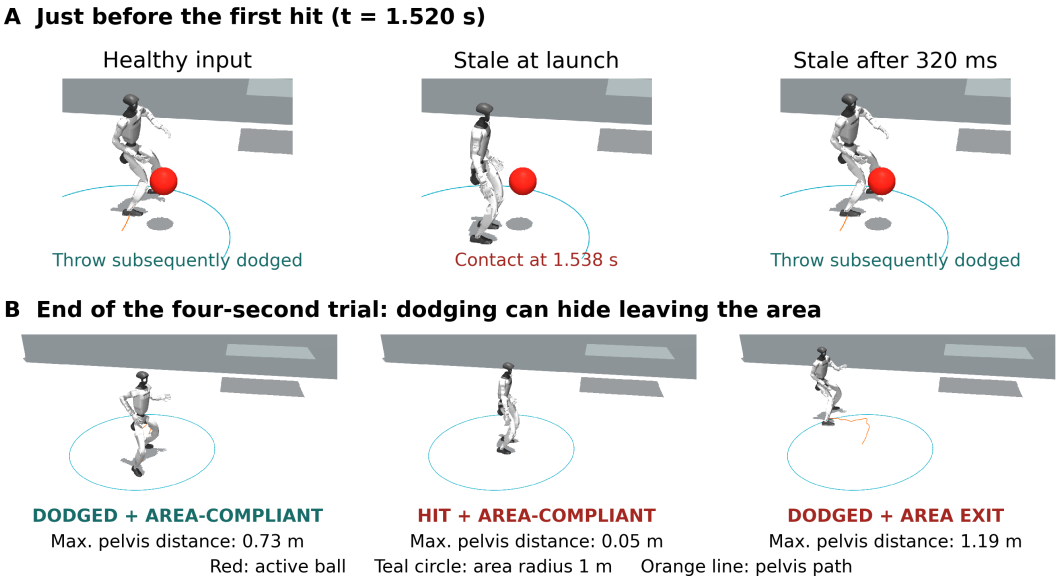


Fig. 1. A dodge can conceal task failure. Exact replays of the first test seed (3000) for π_1, with matched views and colors changed for visibility. Top: just before the middle robot is hit. Bottom: the four-second endpoint. The late stale-input fault allows a dodge but the robot leaves the allowed area. The circle marks the allowed-area boundary (radius 1 m); orange shows the pelvis path. This example illustrates timing, not average performance.

TABLE I
EXPERIMENTAL COHORTS. COUNTS INCLUDE BOTH POLICIES; PILOTS AND PREFLIGHT TRIALS ARE EXCLUDED.

Experiment	Research question	Camera conditions	Matching requirements	Seeds/ policy	Trials
Input handling	How do stale and blank inputs change dodging and area compliance?	Healthy, continuous stale, continuous blank	Same prelaunch robot state and throw within each policy and seed.	192	1,152
Fresh-image restoration	Does restoring fresh input change post-encounter motion?	Healthy, continuous stale or blank, stale or blank → fresh at 2 s	Same prelaunch state and throw; physical traces match before 2 s within each restoration pair.	64	640
Shared-throw control	Do policy responses persist with identical ball launches?	Same five conditions as fresh-image restoration	Same ball launch across both policies and all conditions; robot states may differ.	64	640
Stale-to-blank control	Does removing stale input reproduce the fresh-input success benefit?	Stale until 2 s, then continued stale, blank, or fresh input	Within-policy local triplets: same throw, frame hashes before 2 s, and pelvis traces through 2 s.	64	384

than imposed. This control is analysed separately and adds no independent training runs.

The stale-to-blank control tests whether the success-rate benefit requires fresh frames. We rerun the 64 restoration seeds locally, switching at 2 s from stale input to continued stale input, blank input, or fresh frames. Each triplet matches throws, supplied frame hashes before 2 s, and recorded pelvis traces through 2 s. Although runtime input files match archived hashes, rerun trajectories differ. Every comparison therefore uses the local conditions, separately from the original restoration estimate. Table I summarises the cohorts; Supplement A gives selection and matching details.

C. Outcomes and uncertainty

Task success requires a dodge (a simulator-recorded miss), no fall, and area compliance throughout the trial. The simulator records a fall when pelvis height drops below 0.45 m or upright-axis cosine drops below 0.4 (about 66.4° tilt). Trials that end before the throw is resolved (aborted throws) count as failures and remain included in the total trial count. The area radius is an illustrative task requirement, not an injury threshold or court rule.

Post-hoc timing checks compare the first sampled area exit with throw resolution. A hit resolves at contact. A miss resolves at the first control step at least one second after launch, provided the ball remains active after contact processing and the fall check. Resolution follows the simulator's scoring rule rather than measuring closest approach. An exit is late only if it occurs after resolution; ties count as at-or-before resolution. Samples are 20 ms apart, without interpolation.

Pelvis distance is horizontal distance from its starting position. Maximum and final pelvis distances measure excursion and endpoint distance without applying the area criterion. Post-hoc radius and duration checks reuse all trials, including hits.

Paired 95% confidence intervals use 10,000 bootstrap resamples of evaluation seeds, keeping their conditions together.

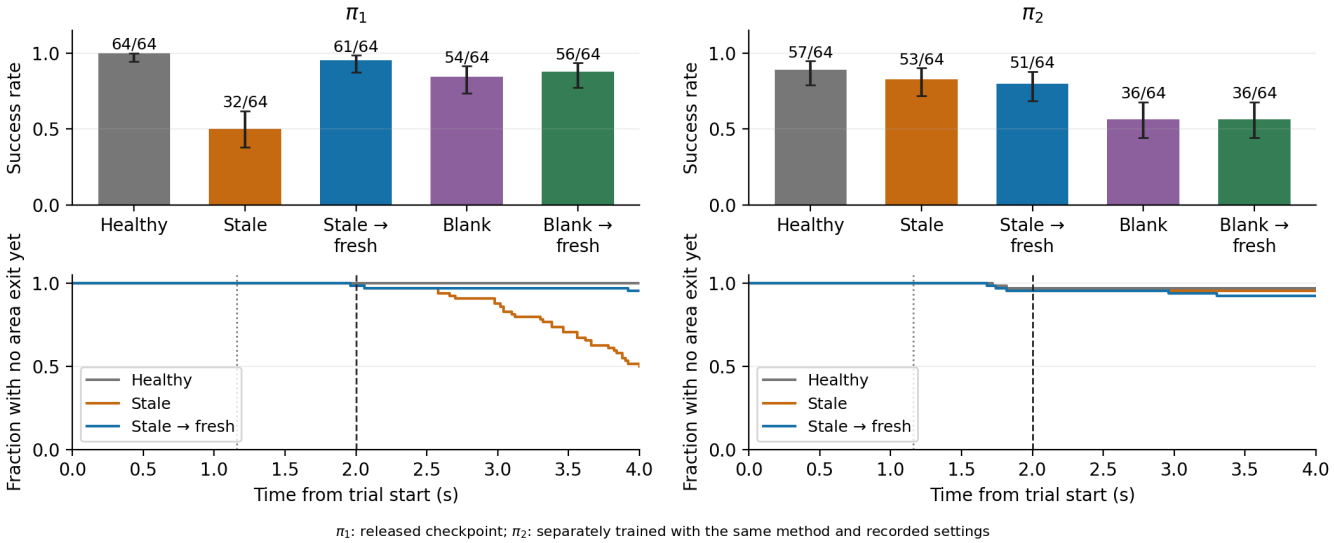


Fig. 2. Original restoration experiment, all 640 trials. Top: success rates with Wilson 95% confidence intervals and counts out of 64. Bottom: fraction with no area exit yet; an exit remains counted even if the robot returns. Dotted lines mark fault onset at 1.16 s; dashed lines mark fresh-image restoration at 2 s. Throws can differ across policies.

Intervals are per comparison without multiplicity adjustment; individual-condition intervals are Wilson intervals. Evaluation seeds are not independent training runs. Supplement A gives scoring rules and Supplement B reports additional statistics.

III. Results

A. Dodges can hide late area exits

Under stale input, π_1 dodges successfully in 121 trials but later leaves the allowed area. All 122 of its area exits occur after throw resolution. Blank input also produces late exits, as does π_2. These timings establish when area-compliance failures occur, not why the policies differ (Supplement B).

B. Fresh frames change post-encounter motion

Restoring fresh frames raises π_1 success from 32/64 to 61/64 (+45.3 percentage points; paired 95% confidence interval +32.8 to +57.8), while both conditions dodge 64/64 throws. This rescues 29 trials, with none changing from success to failure. For π_2, success changes from 53/64 to 51/64 (-3.1 percentage points; paired 95% confidence interval -7.8 to 0), with 54/64 dodges in both conditions. No trials improve and two change from success to failure. No restoration condition has falls or aborted throws. The success differences therefore concern area compliance after the encounter, without rescuing the initial dodge. Table II reports all conditions; Fig. 2 shows when exits occur.

C. Success-rate increases depend on area radius

The increase in success rate emerges mainly late in the trial and shrinks with larger area radii. At four seconds, increasing the radius from 1 to 1.5 m reduces π_1's additional successful trials from 29/64 to 2/64 in the original cohort. Yet fresh frames reduce its mean maximum pelvis distance from 1.043 to 0.860 m, independently of that criterion. For π_2, mean final pelvis distance decreases, but mean maximum pelvis distance increases slightly in the original cohort. The shared-throw confidence interval for the change in mean maximum pelvis distance spans zero. Supplement B gives post-hoc methods and selected results.

D. Blank input has different effects across policies

Compared with stale input, blank input raises π_1 success from 70/192 to 171/192 but lowers π_2 success from 158/192 to 100/192. Both policies stay in the area more often but dodge less often (Table III). These outcomes account for the opposite success-rate changes without explaining their cause inside the policies.

The response difference persists under shared ball launches. Fresh-image restoration also raises π_1 success from 23/64 to 60/64 in that control, while π_2 changes from 46/64 to 45/64; dodge counts remain unchanged. However, the shared-throw confidence interval for π_2's blank-versus-stale effect reaches zero, so this control alone does not establish a harmful blank-input effect for that policy (Supplement B).

E. Equal success can hide different final distances

In the local 2 s switch control, fresh and blank input produce identical success outcomes but different final positions. Fresh frames bring both policies closer to their starting position than blank input does. Mean final pelvis distance is lower by 0.295 m for π_1 (paired 95% confidence interval 0.253–0.341 m) and 0.117 m for π_2 (0.091–0.144 m). The same success outcomes can thus conceal different post-encounter motion.

This endpoint improvement does not establish a reduction in maximum excursion. For both policies, the paired 95% confidence interval for the fresh-versus-blank difference in mean maximum pelvis distance spans zero. These distance summaries include all 64 trials per policy, including hits.

TABLE II
ORIGINAL RESTORATION EXPERIMENT. COUNTS OUT OF 64 PER CONDITION; FALLS AND ABORTED THROWS ARE ZERO.

Policy	Camera condition	Dodged	Area-compliant	Successful
π_1	Healthy	64	64	64
	Stale	64	32	32
	Stale $\rightarrow$ fresh	64	61	61
	Blank	60	57	54
	Blank $\rightarrow$ fresh	60	60	56
π_2	Healthy	57	62	57
	Stale	54	61	53
	Stale $\rightarrow$ fresh	54	59	51
	Blank	36	63	36
	Blank $\rightarrow$ fresh	36	63	36

TABLE III
INPUT-HANDLING EXPERIMENT. COUNTS OUT OF 192 PER CONDITION. SUCCESS REQUIRES A DODGE, NO FALL, AND AREA COMPLIANCE.

Policy	Camera condition	Dodged	Fell	Area-compliant	Successful
π_1	Healthy	192	0	192	192
	Stale	191	0	70	70
	Blank	182	0	181	171
π_2	Healthy	162	0	189	161
	Stale	165	0	182	158
	Blank	101	0	190	100

Supplement B gives the distance statistics; the next subsection examines the success outcomes and what the blank-input comparison tests.

F. Blank frames reproduce the observed success outcomes

Restoring fresh frames both supplies current visual information and stops repeating the stale frame. The stale-to-blank control separates these changes by replacing the stale frame with zeros at 2 s, after throw resolution, while retaining the policy histories. It tests whether fresh images are needed for the observed success-rate benefit at area radius 1 m.

In this separate local control, π_1 succeeds in 31/64 trials with continued stale input and 62/64 with either blank input or fresh frames. Both switches rescue the same 31 trials, with none changing from success to failure. All 64 throws are dodged under every condition, so these gains come entirely from preventing area exits. For π_2, all three conditions succeed on the same 54/64 trials and dodge 55/64 throws: neither switch changes success. No falls or aborted throws occur. Blank input thus reproduces the observed success-rate benefit for π_1 and the absence of a benefit for π_2. Identical outcomes on these seeds do not establish population equivalence. Supplement B gives all counts and the discordance bound.

The control shows that fresh visual information is not required to reproduce the observed binary success gain in this setting, although it improves mean final pelvis distance. It does not show that the policies ignore vision: histories retain earlier information, and blank frames still pass through the visual encoder. Nor does it identify the mechanism inside either policy.

IV. RELATED WORK AND LIMITATIONS

Our policies remain fixed during testing. Their released training already includes image corruption and repeated frames [1]. VB-Com switches between vision and blind policies [8], while CAP trains robustness to degraded depth [3]. Supplying blank input does not turn a visual model into a separately trained blind policy. OopsieVerse distinguishes completion from damage-free completion [2]; our area criterion measures task compliance rather than damage. Like LIBERO-MAX's matched execution-time interventions [14], our comparisons hold the preceding trial history fixed within each restoration pair.

The late-switch finding does not establish blank input as a general fallback. The input-handling experiment supplies blank input from 1.16 s, during the encounter, whereas the stale-to-blank control switches at 2 s, after throw resolution. The experiments also use different cohorts and execution environments. Their results therefore do not provide a paired estimate of the effect of switching time.

The evidence covers two fixed policies, one simulated robot and starting posture, the native throw generator, and one recorded launch pool. It does not establish hardware safety, human/team interaction performance, a universal fallback, or reliability across training seeds. Pelvis containment does not ensure whole-body containment. The hard area criterion is additional to the smooth station-hold training reward, which has weight 0.3 and scale 1 m (Supplement A). The reward scale does not itself justify that hard criterion. Training variability is well documented [4], but our models do not isolate it, and fixed hyperparameters leave their sensitivity untested. Further work should vary checkpoints, stance and pre-impact configuration, and test actuator limits and latency on hardware. Original archives and π_2 ONNX weights are public; stale-to-blank records accompany this revision and await public release (Supplement C).

V. CONCLUSION

A successful dodge can hide a later area-compliance failure. Restoring fresh frames raises π_1 success from 32/64 to 61/64 without changing its dodge count, showing why dodge rate alone is insufficient. In the separate stale-to-blank control, blank input and fresh-image restoration produce the same success outcomes, yet fresh frames bring both policies closer to their starting position by trial end. Overall success can therefore also conceal differences in motion. These findings from two fixed policies in simulation support reporting post-encounter motion alongside contact, falls, and area compliance.

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